

**PCB DESIGN PROJECT**

# **Design of Battery Low Voltage Cut-Off Circuit Using Transistor**

Project Documentation Report

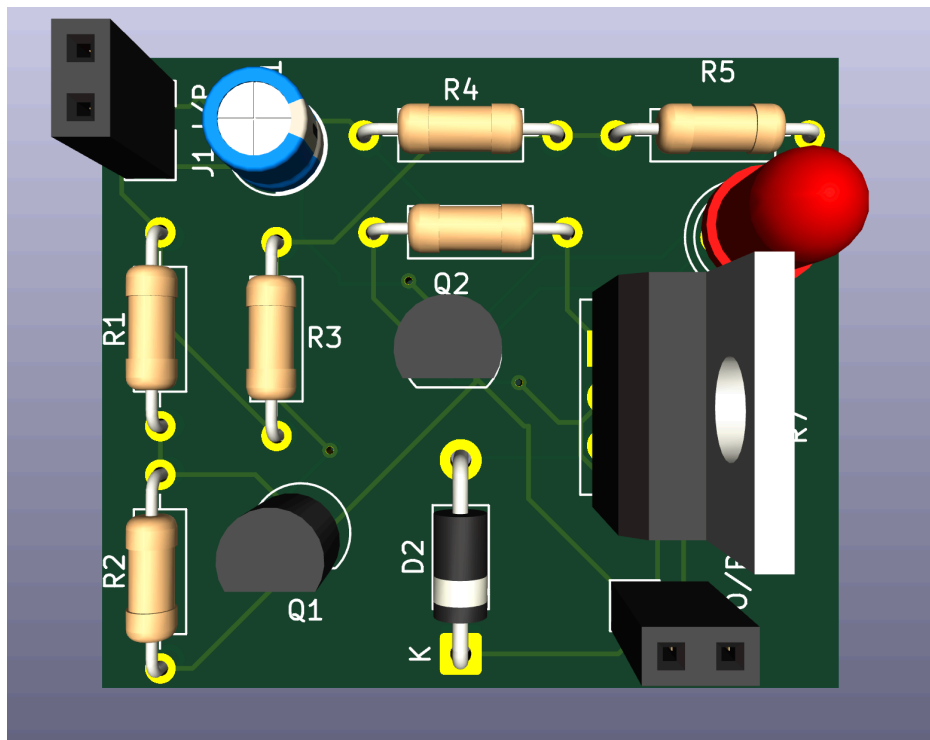
Submission Task 7

|               |                                                      |
|---------------|------------------------------------------------------|
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| College       | Jspm Rajarshi Shahu college of engineering           |
| Branch        | Electronics And Telecommunication                    |
| Project       | Battery Low Voltage Cut-Off Circuit Using Transistor |
| PCB           | Two-layer Through-Hole (THT) design                  |
| Software      | eSim / eSchema / KiCad PCB Editor                    |
| Main function | Detect low battery voltage and disconnect the load   |

## 1. Project Overview

The aim of this project is to design a simple and practical battery low-voltage cut-off circuit using discrete transistors and a power MOSFET. The circuit continuously senses the battery voltage and disconnects the load when the voltage falls below the selected threshold. This type of protection is useful in battery-powered systems because it prevents the load from continuing to operate when the battery reaches an undesirably low voltage.

The design was developed as a complete PCB project rather than only as a schematic. The work includes the circuit schematic, component footprint assignment, two-layer PCB placement, routing, design-rule checking and 3D inspection. Through-hole components were selected because they are easy to place, inspect and assemble and are suitable for a practical academic PCB design.

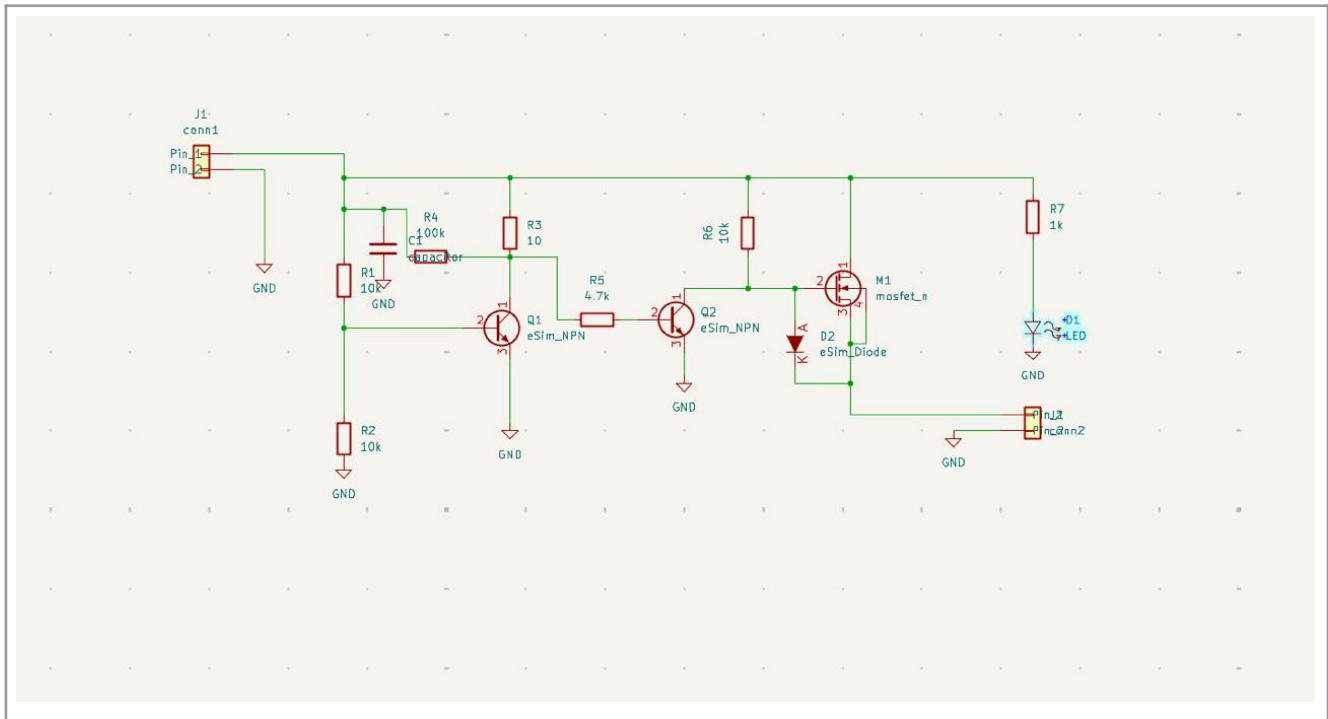


## 2. Circuit Description

The battery is connected at the input through J1. R1 and R2 form a voltage divider that reduces the battery voltage to a sensing level suitable for the transistor threshold stage. The junction of R1 and R2 is the SENSE node and is connected to the base of Q1. Q1 is an NPN transistor used to detect the change in the sensed voltage. R3 acts as the collector pull-up resistor and produces the control signal used by the next transistor stage.

The control signal from Q1 is connected to Q2 through R5. Q2 acts as the gate-driver stage for the power MOSFET M1/Q3. The MOSFET is responsible for switching the load connection. R6 provides the gate pull-up. R4 is used as the feedback resistor for hysteresis, which helps reduce unwanted rapid switching when the battery voltage is close to the cut-off point. C1 is connected to the sensing node to reduce noise and improve the stability of the sensing signal.

At the output, J2 provides the connection to the load. LED D1 together with R7 provides a visual indication of the load-connected state. The protection diode D2 is placed around the MOSFET gate/source section according to the selected circuit arrangement. The exact polarity and device model should be verified against the final schematic and selected MOSFET before hardware fabrication.



### 3. Schematic

The schematic was arranged in functional sections so that the signal flow can be followed from left to right: battery input, voltage sensing, threshold detection, gate driving, MOSFET switching and load indication. The main nets used in the design are VBAT, SENSE, GATE, LOAD and GND. This arrangement makes the circuit easier to troubleshoot and also provides a clear basis for the PCB placement.

### 4. Footprint Assignment

The project uses THT footprints for the physical PCB. R1 to R7 use standard axial resistor footprints. Q1 and Q2 use TO-92 footprints suitable for the selected NPN transistors, while M1/Q3 uses a TO-220 style footprint for the power MOSFET. D1 uses a 5 mm through-hole LED footprint and D2 uses a suitable through-hole diode footprint. C1 uses a radial THT capacitor footprint.

The battery and load connections are made using connectors, so the connection points can be implemented using simple through-hole pin/header or test-point style footprints rather than large terminal blocks. The final footprint selection must match the physical component and, especially for transistors and MOSFETs, the electrical pin arrangement. A TO-92 or TO-220 package name alone does not guarantee the same pin order for every device.

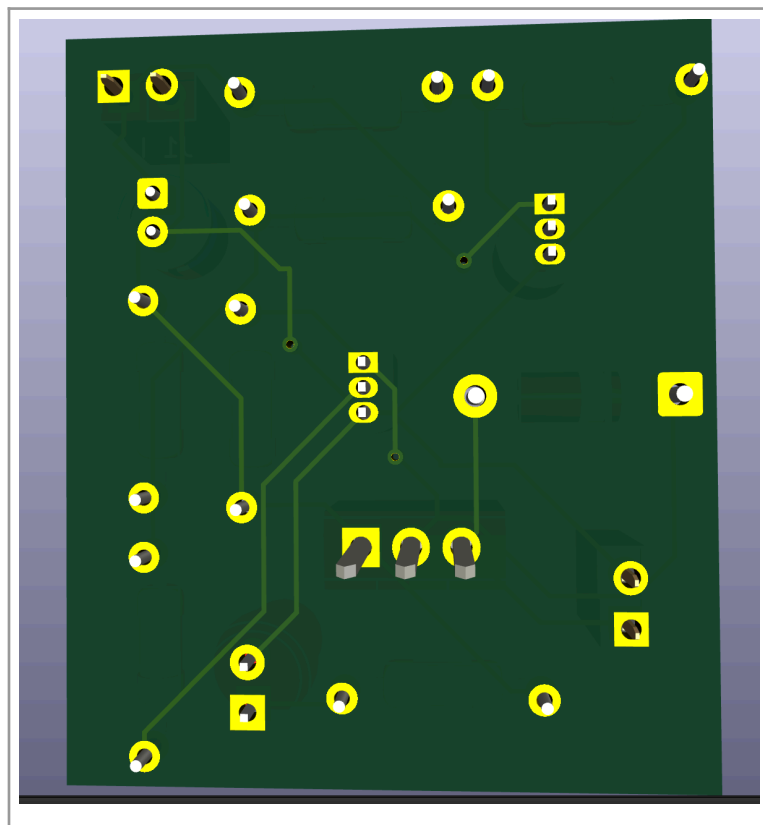
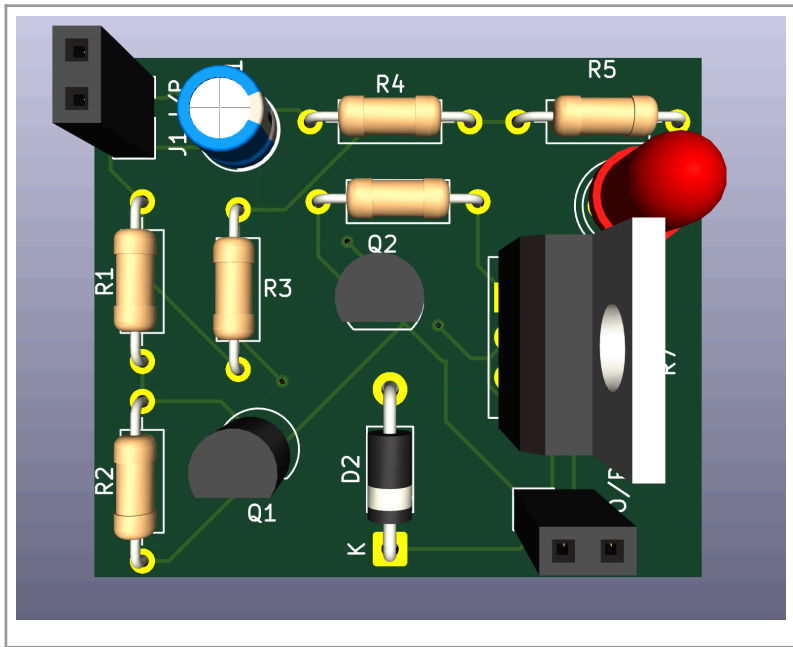
#### Symbol : Footprint Assignments

|    |      |                                                                       |
|----|------|-----------------------------------------------------------------------|
| 1  | C1 - | capacitor : Capacitor_THT:CP_Radial_D5.0mm_P2.00mm                    |
| 2  | D1 - | LED : LED_THT:LED_D5.0mm                                              |
| 3  | D2 - | eSim_Diode : Diode_THT:D_DO-41_SOD81_P10.16mm_Horizontal              |
| 4  | J1 - | conn1 : Connector_PinSocket_2.54mm:PinSocket_1x02_P2.54mm_Vertical    |
| 5  | J2 - | conn2 : Connector_PinSocket_2.54mm:PinSocket_1x02_P2.54mm_Vertical    |
| 6  | M1 - | mosfet_n : Package_TO_SOT_THT:TO-220-3_Vertical                       |
| 7  | Q1 - | eSim_NPN : Package_TO_SOT_THT:TO-92_Inline                            |
| 8  | Q2 - | eSim_NPN : Package_TO_SOT_THT:TO-92_Inline                            |
| 9  | R1 - | 10k : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal  |
| 10 | R2 - | 10k : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal  |
| 11 | R3 - | 10 : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal   |
| 12 | R4 - | 100k : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal |
| 13 | R5 - | 4.7k : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal |
| 14 | R6 - | 10k : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal  |
| 15 | R7 - | 1k : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal   |

## 5. PCB Layout

The PCB was designed as a two-layer board using the front and bottom copper layers. Component placement was arranged according to the function of each circuit section. The input, MOSFET and output connection are kept relatively close so that the main battery-to-load path remains short. The sensing components are grouped together to keep the low-voltage sensing network compact, while Q2, R5 and R6 are placed close to the MOSFET gate to reduce the gate-drive path.

The LED and other user-accessible components are positioned so that they can be easily observed or operated. The layout also leaves enough spacing around the THT components for soldering and inspection. The board outline defines the physical PCB boundary and all components are placed within that boundary.

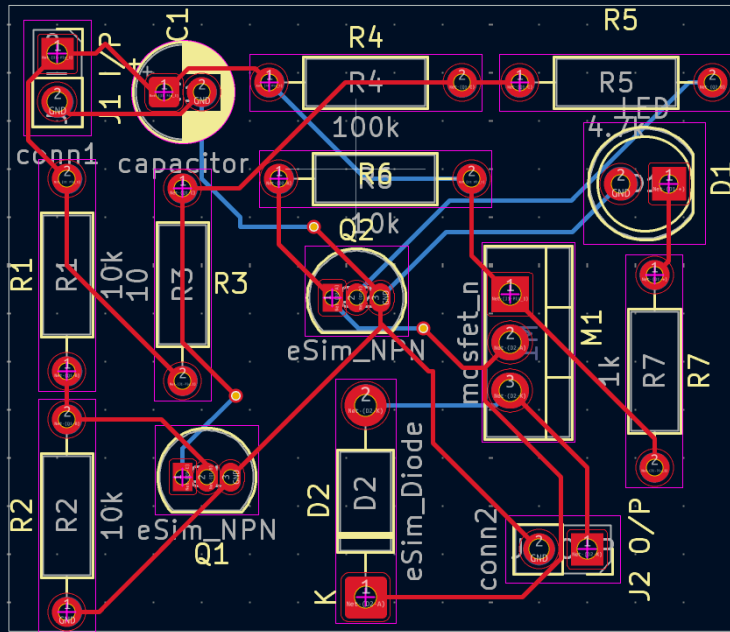


## 6. Routing Methodology

The routing was planned by separating the power path from the sensing and control paths. The battery and load connections carry the main current through the MOSFET, so these traces should be wider than the low-current signal traces. The R1/R2 sensing node is kept short because it is the most sensitive control

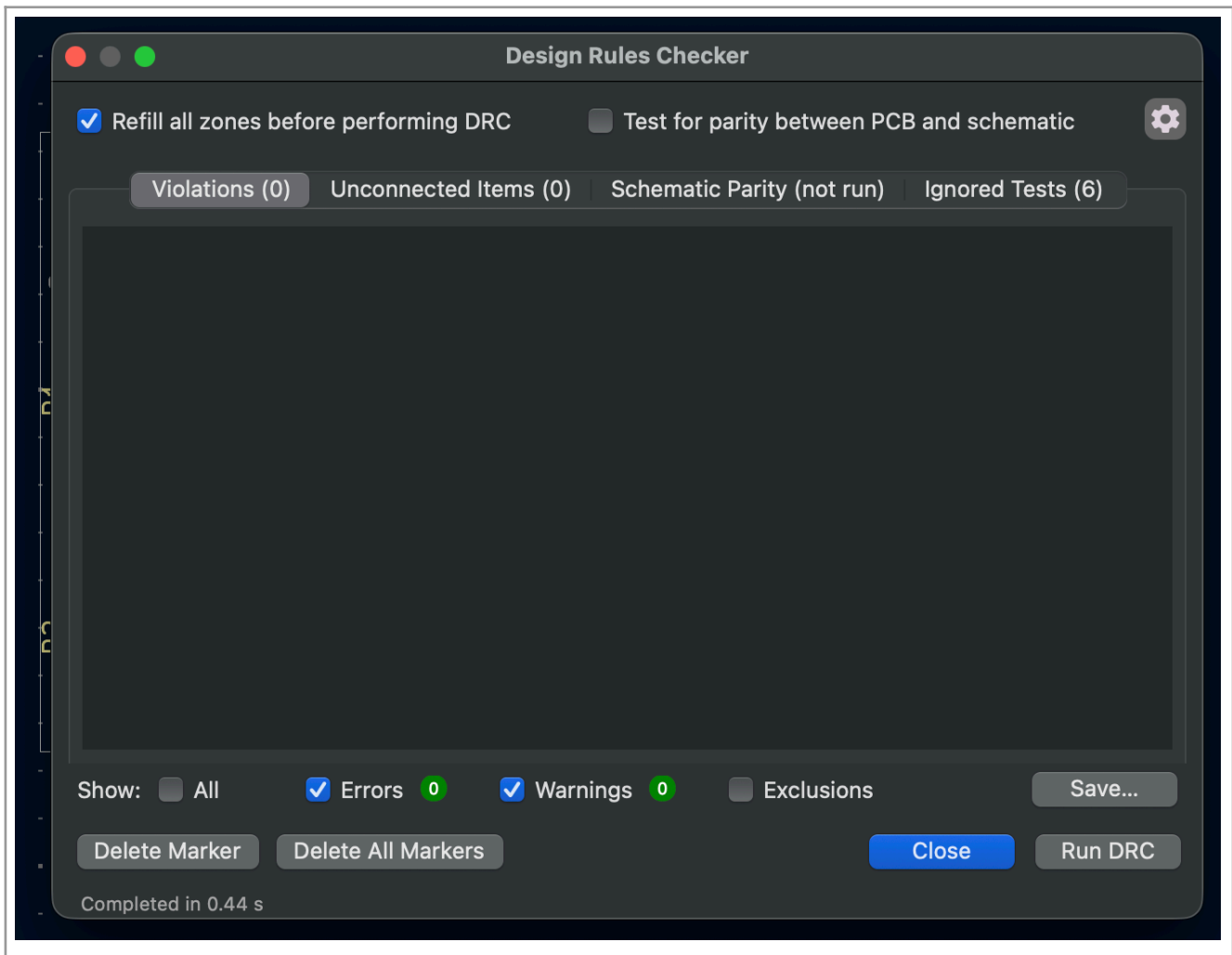
node in the circuit. The connection between Q2 and the MOSFET gate is also kept short to improve the switching path.

The two copper layers are used to complete the connections while keeping the routing organized. The bottom layer can be used for ground return and a ground zone where appropriate. The routing is arranged to avoid unnecessary long tracks and sharp corners, and the final layout should be checked for clearance and unrouted connections before DRC.



## 7. DRC Results

The Design Rules Checker is used after routing to identify PCB layout problems such as clearance violations, unrouted nets, overlapping footprints and other rule violations. The DRC result shown in the final submission should represent the actual state of the completed PCB. Any critical errors must be corrected before the project is submitted.



## 8. Design Considerations

A major consideration in this design is the separation between the sensing section and the power-switching section. The sensing network works with relatively small voltages and can be affected by noise from the high-current path, so its traces and components should be kept compact and away from the main power routing. The MOSFET should be placed close to the input and output connections, and the power traces should be selected according to the expected load current.

The cut-off voltage is determined by the actual resistor values and transistor behavior used in the circuit. It should therefore be verified through the intended simulation or hardware test rather than treated as a universal value. The circuit is a low-voltage cut-off stage and is not, by itself, a complete battery-management system; additional protection functions may be required depending on the battery chemistry and application.

## 11. Conclusion

The Battery Low Voltage Cut-Off Circuit Using Transistor provides a compact and practical example of a battery protection stage implemented as a PCB. The design combines voltage sensing, transistor-based

threshold detection, hysteresis, MOSFET load switching and LED indication in a two-layer THT board. The project also demonstrates the complete PCB development process, starting from the schematic and footprint assignment and continuing through placement, routing, DRC and 3D verification.

The final PCB should be considered complete only after the schematic connectivity, component pin mapping, board outline, routing, DRC results and 3D view have all been checked. The documentation should contain the actual screenshots from the completed project so that the report directly reflects the submitted design.

## 12. Submission Files

The final GitHub submission should contain the complete project rather than only a PCB image. The recommended package includes the eSim/KiCad project file, schematic file, PCB layout file, supporting libraries or configuration files where required, README.md, this documentation report and the final verification screenshots. Keeping the complete project together allows the design to be reopened and reviewed without missing project information.

| File / Item            | Purpose                                    |
|------------------------|--------------------------------------------|
| Schematic (.kicad_sch) | Complete circuit design                    |
| PCB (.kicad_pcb)       | Final two-layer PCB layout                 |
| Project (.kicad_pro)   | Project configuration                      |
| README.md              | Project summary and repository information |
| Documentation report   | Detailed project explanation               |
| ERC/DRC screenshots    | Design verification                        |
| PCB/3D screenshots     | Visual documentation                       |